

— DOMAIN 1 · FOUNDATIONS —

# Foundations of Accounting

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The one record behind every cost report, invoice and forecast — built from first principles, for the project-controls professional.

5 Knowledge Areas   ~55 min lecture   Certified Project Controls Professional — AI

— WHY THIS MATTERS TO PROJECT CONTROLS

# Every number you report is a view into one record

Cost reports, EV curves, invoice applications, forecasts — all of them draw, ultimately, on the same double-entry ledger. Domain 1 builds that record so the figures stop being a black box.

KA 1.1

## The accounting model

Equation, debits & credits, journal → ledger → trial balance.

KA 1.2

## The statements

The four financial statements and how they lock together.

KA 1.3

## Accrual & matching

Recording economic events, not cash movements.

KA 1.4

## Provisions & accruals

Cost you know is coming — the IAS 37 judgement.

KA 1.5

## Chart of accounts

Cost coding, the CBS and mapping to the WBS.

# 1.1

## The accounting model

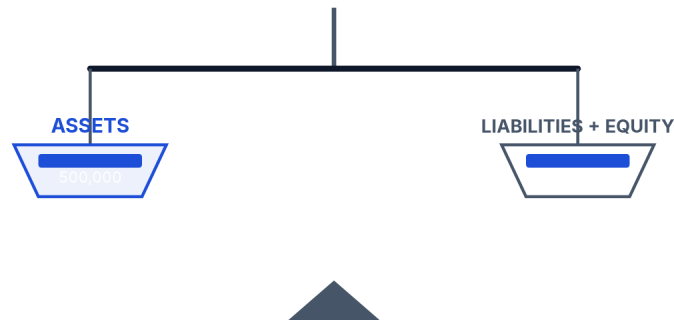
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By the end you will be able to:

- 1 State the accounting equation and explain why every transaction keeps it in balance.
- 2 Apply debit and credit rules across the five account types.
- 3 Follow one transaction cycle: journal → ledger → trial balance → equation.
- 4 Explain what a trial balance proves — and, crucially, what it cannot.

# Assets = Liabilities + Equity

The identity underlying the IFRS Conceptual Framework — true for a company, a joint venture, or a single-project vehicle. It holds because of **double aspect**: every event is recorded twice, once on each side of the scale.

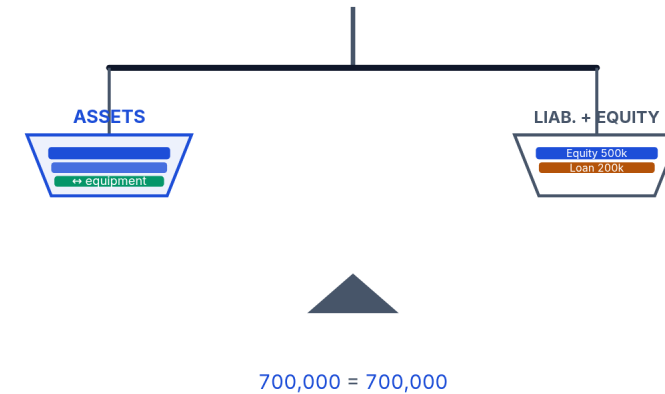


**Double aspect.** Borrow cash → an asset *and* a liability rise together. Buy kit for cash → one asset falls, another rises. The scale never tips.

# Three transactions, one level scale

| DAY-ONE EVENTS                     | USD     |
|------------------------------------|---------|
| 1 · Share capital paid in cash     | 500,000 |
| 2 · Bank loan drawn in cash        | 200,000 |
| 3 · Site equipment bought for cash | 150,000 |
| Total assets                       | 700,000 |

**The classic trap.** Transaction 3 swaps cash for equipment — assets' *composition* changes, the *total* doesn't. Not every entry hits profit.



# Five account types, one rule each

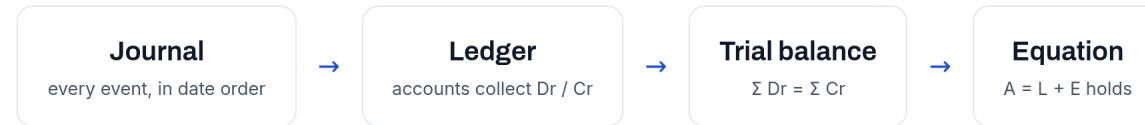
**Debit** (Dr, left) and **credit** (Cr, right) carry no moral meaning — they're a labelling system. What matters is each account type's **normal balance**: the side on which it increases.

| ACCOUNT TYPE | INCREASES WITH | NORMAL BALANCE | EXAMPLES                          |
|--------------|----------------|----------------|-----------------------------------|
| Asset        | Debit          | Debit          | Cash, receivables, equipment      |
| Expense      | Debit          | Debit          | Labour, materials, depreciation   |
| Liability    | Credit         | Credit         | Loans, payables, provisions       |
| Equity       | Credit         | Credit         | Share capital, retained earnings  |
| Income       | Credit         | Credit         | Contract revenue, interest income |

**Hold it as one idea:** uses of resources (assets, expenses) are debit-normal; sources (liabilities, equity, income) are credit-normal. And every entry obeys  $\sum \text{Dr} = \sum \text{Cr}$ .

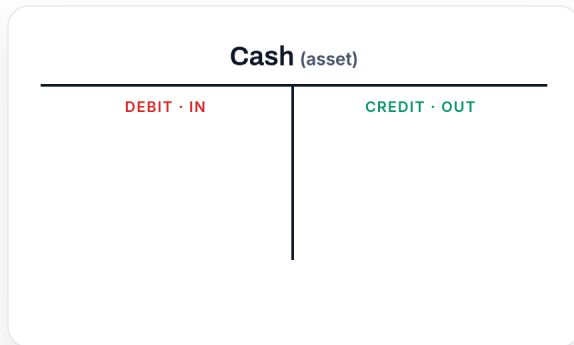
# Journal → ledger → trial balance → equation

Four steps turn raw events into checked figures. Capture first, classify second, prove third, close fourth — the same mechanism for a single-project vehicle or a multinational.



Meridian records **11 transactions** in March 2026 — capital, a loan, equipment, materials, wages, a subcontractor, a client invoice, a receipt, a supplier payment and depreciation. Watch one account fill.

# The Cash T-account, filling



Posting is **purely mechanical** once the entry is classified: each line lands in its named account, tagged with a reference back to the journal.

**Control accounts.** Where volume is high, a subsidiary ledger holds supplier-level detail and a single control account carries the total — the two must reconcile every period.



## Both columns prove equal

Every ledger balance, split Dr / Cr. If the two totals agree, the double-entry was applied consistently across the whole ledger.



Total debits

**\$1,022,500**



Total credits

**\$1,022,500**

Equal, to the dollar — the books are **in balance**.

# What a balanced trial balance cannot catch

It proves only that  $\sum Dr = \sum Cr$ . Any error that preserves that identity slips straight through.

## Error of principle

Right amount, wrong account of the *same* normal side (materials → labour). Totals unchanged.

## Omission

A transaction never recorded at all. Both sides simply lower together.

## Commission

Posted to the wrong account of the same type — say, the wrong supplier.

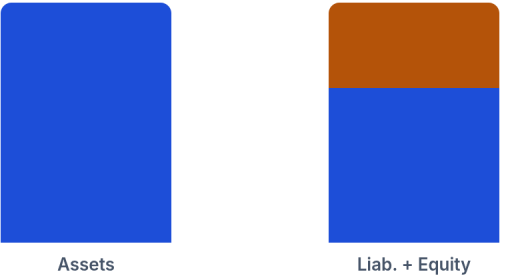
## Compensating

Two unrelated errors that happen to offset each other exactly.

**Exam-critical:** a balanced trial balance is **necessary but not sufficient**. In project controls these errors surface not from the trial balance but from cost-code and WBS reconciliation (KA 1.5).

# Profit becomes equity; the equation holds

| ROLL-UP, MARCH 2026      | USD       |
|--------------------------|-----------|
| Contract revenue         | 220,000   |
| Less all costs           | (177,500) |
| Net profit               | 42,500    |
| Opening equity (capital) | 500,000   |
| Closing equity           | 542,500   |



842,500      300k + 542.5k

$842,500 = 300,000 + 542,500 \checkmark$

— KA 1.1 · KNOWLEDGE CHECK

**A trial balance balances, yet a \$15,000 subcontract cost was posted in full to “Labour cost.” Why did the trial balance *not* catch it?**

- ☐ A The trial balance only checks liabilities, not expenses.
- ☒ B Both accounts are debit-normal expenses, so total debits were unchanged.
- ☐ C It caused an equal, opposite understatement of accounts payable.
- ☐ D A trial balance can't be prepared when such an error exists.

**B is correct.** Moving an amount between two accounts on the *same* normal side leaves  $\Sigma \text{Dr} = \Sigma \text{Cr}$  untouched — an error of principle the trial balance is blind to. It's found by account-level or cost-code review, not the trial balance.

# 1.2

## Components of the financial statements

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- 1 Name the four primary statements and the question each answers.
- 2 Read a Statement of Financial Position as the accounting equation, frozen in time.
- 3 Explain how the statements **articulate** — lock together into one coherent picture.

# Four statements, four questions

IAS 1

## Statement of Financial Position

What we **own and owe** at a point in time — the equation, frozen.

IAS 1

## Profit or Loss & OCI

How we **performed** over the period — income less expenses.

IAS 7

## Cash Flows

Where **cash** moved — operating, investing, financing.  
Profit  $\neq$  cash.

IAS 1

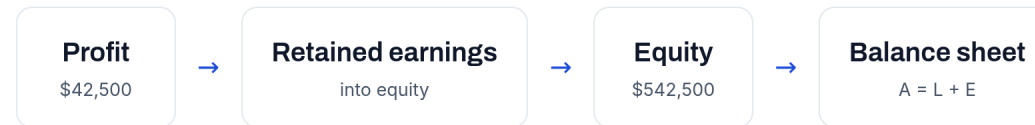
## Changes in Equity

How the **owners' stake** moved — profit, and distributions.

**Plus the notes.** Basis of preparation, accounting policies and the judgements behind the numbers — where the real story of a construction contract often lives.

# They lock together

The statements share figures by design — this is why a set that “articulates” is internally consistent, and why an error in one surfaces in another.



And separately: the **cash flow** statement's closing figure — Meridian's **\$605,000** — is the cash line on the balance sheet. Same number, two windows.

# 1.3

## **Accrual accounting & matching**

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- 1 Contrast the cash basis with the accrual basis.
- 2 Apply the matching principle — cost recognised with the revenue it earns.
- 3 Post an accrual for work done but not yet invoiced, and reverse it.



— KA 1.3 · TWO BASES

# When do you record it?

CASH BASIS

## When money moves

Record on receipt or payment. Simple — but it distorts performance: an unpaid subcontractor bill makes a bad month look good.

ACCRUAL BASIS · REQUIRED BY IFRS

## When the event occurs

Record the cost when work is done and the revenue when it's earned — whatever the cash is doing. The only basis that gives a true cost report.

**The matching principle** ties them: recognise a cost *in the same period* as the revenue it helped earn — not when its invoice happens to arrive.

# Cost done, invoice not yet arrived

Month-end: Meridian’s subcontractor has completed \$18,000 of verified work but has not invoiced. Cash basis records nothing — and understates the month.

**Why controls care:** value of work done, not cash paid, is what earned value and the cost report must reflect. Missing accruals are the classic cause of a “great month” that reverses.

PERIOD-END ADJUSTING ENTRY

|                     |        |
|---------------------|--------|
| Dr Subcontract cost | 18,000 |
| Cr Accrued expenses | 18,000 |

REVERSED ON DAY 1 NEXT PERIOD

|                     |        |
|---------------------|--------|
| Dr Accrued expenses | 18,000 |
| Cr Subcontract cost | 18,000 |

...so the real invoice, when it lands, isn't counted twice.

# 1.4

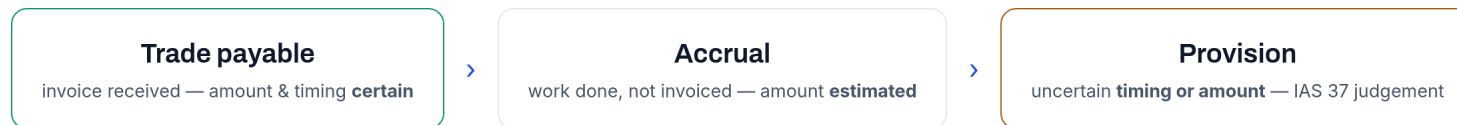
## Cost provisions & accruals

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- 1 Place an item on the certainty spectrum: payable → accrual → provision.
- 2 Apply the IAS 37 three-part recognition test.
- 3 Recognise a contract-loss (onerous-contract) provision — a real project risk.

# From invoice to informed estimate

The same idea — a cost the entity owes — at three levels of certainty. As certainty falls, judgement rises.



**IAS 37 defines a provision** as “a liability of uncertain timing or amount.” The question is: when are you *allowed* to recognise one?

# Three gates — all must open

1

## Present obligation

A legal or constructive duty **from a past event**. No past event → stop.

2

## Probable outflow

Settlement will **probably** require resources (more likely than not).

3

## Reliable estimate

The amount can be **reliably estimated**.

✓ **Recognise a provision**

Fail gate 2 → **contingent liability** (disclose in the notes, don't book). Fail gate 1 → nothing at all.

# Warranty, and the loss-making contract

## WARRANTY PROVISION

### Expected value, discounted

Post-handover rectification is probable and estimable. Measure at expected value; discount for timing → e.g. \$95,238 today.

## ONEROUS CONTRACT

### Provide the whole loss, now

When unavoidable costs to finish **exceed** revenue left to earn, the full expected loss is provided for **immediately**. You can't spread bad news.

**AI in this topic.** Models can flag a contract trending onerous from the cost forecast — a genuine early-warning use. But recognising the provision is a judgement the professional owns. *AI proposes; the professional disposes.*

# 1.5

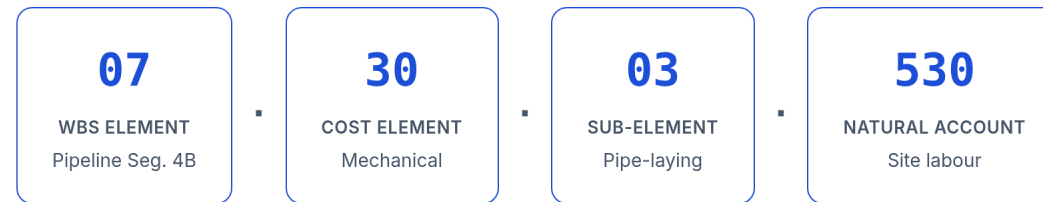
## Chart of accounts & cost coding

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- 1 Read a project cost code and what each segment means.
- 2 Connect the cost breakdown structure to the WBS and the control account.
- 3 Judge where AI genuinely helps with coding — and where it must not decide.

# One code, many views

Each segment is an address. Together they let a single ledger line be sliced by work package, cost type *and* natural account.



**Control account = WBS × OBS.** Where a slice of work meets the team accountable for it, you get the control account — the cell where scope, cost and schedule are actually managed, and where cost coding, the CBS and the WBS all reconcile.



# Where AI helps — and where it stops

## GENUINELY USEFUL

### Auto-classify & flag

Suggest a cost code from past patterns; surface anomalies — a labour code with no timesheet hours (the very error the trial balance missed).

## THE GOVERNANCE LIMIT

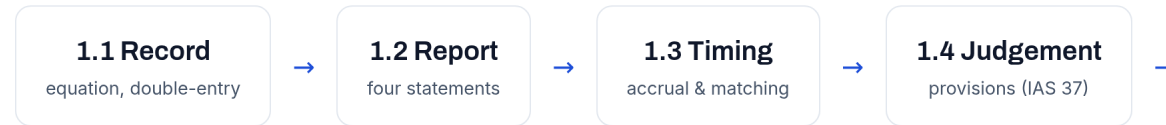
### It doesn't know *why*

It learns past mistakes as readily as good practice, and can't explain a break. Material or novel items route to a human.

**The principle of the whole programme:** *AI proposes; the professional disposes.*

— DOMAIN 1 · THE THROUGH-LINE

# One connected system



— END OF DOMAIN 1 —

# “AI proposes; the professional disposes.”

Next — **Domain 2: Financial Reporting & the Standards.** The foundations you’ve just built meet IFRS in depth, including revenue recognition on construction contracts under IFRS 15.